

QueryLib_for_PATLIBs

January 19, 2026

1 PATSTAT A Query Library for PATLIBS

1.1 Setup

```
[2]: from epo.tipdata.patstat import PatstatClient
patstat = PatstatClient(env='PROD')
import pandas as pd
import time

def timed_query(query):
    start = time.time()
    res = patstat.sql_query(query, use_legacy_sql=False)
    print(f"Query took {time.time() - start:.2f}s ({len(res)} rows)")
    return pd.DataFrame(res)
```

1.2 Business Question 1

Which countries have the highest patent application activity since 2015, and what are their grant rates? This helps identify leading innovation hubs and their success rates in obtaining patents.

```
[4]: df = timed_query("""
SELECT p.person_ctype_code,
       COUNT(DISTINCT a.appln_id) AS patent_count,
       COUNT(DISTINCT CASE WHEN a.granted = 'Y' THEN a.appln_id END) AS
↪granted_count,
       ROUND(COUNT(DISTINCT CASE WHEN a.granted = 'Y' THEN a.appln_id END) *
↪100.0 /
              COUNT(DISTINCT a.appln_id), 2) AS grant_rate
FROM tls207_pers_appln pa
JOIN tls206_person p ON pa.person_id = p.person_id
JOIN tls201_appln a ON pa.appln_id = a.appln_id
WHERE pa.applt_seq_nr > 0
      AND a.appln_filing_year >= 2015
      AND p.person_ctype_code IS NOT NULL
GROUP BY p.person_ctype_code
HAVING COUNT(DISTINCT a.appln_id) >= 100
ORDER BY patent_count DESC
LIMIT 20
```

```
""")
df
```

Query took 0.53s (20 rows)

```
[4]:
```

	person_ctry_code	patent_count	granted_count	grant_rate
0		36951698	26019319	70.41
1	US	3999448	1992836	49.83
2	KR	1968836	1107122	56.23
3	JP	1708237	702885	41.15
4	CN	1265833	437470	34.56
5	DE	1204752	482439	40.04
6	TW	533026	428967	80.48
7	FR	457915	222595	48.61
8	GB	331191	138779	41.90
9	CH	300883	122765	40.80
10	NL	208716	94585	45.32
11	IT	207519	73643	35.49
12	CA	205351	96607	47.04
13	SE	175587	79217	45.12
14	RU	131754	117961	89.53
15	IL	104396	39502	37.84
16	AU	97788	38833	39.71
17	ES	95036	52510	55.25
18	AT	89601	44873	50.08
19	FI	86860	36409	41.92

1.3 Business Question 2

What are the most active technology fields in recent years (2018-2022), and how do they compare in terms of patent family size and citation impact? This reveals trending technology areas and their relative importance.

```
[28]: df = timed_query("""
SELECT
    tf.techn_sector,
    tf.techn_field,
    COUNT(DISTINCT a.appln_id) AS application_count,
    AVG(a.docdb_family_size) AS avg_family_size,
    AVG(a.nb_citing_docdb_fam) AS avg_citations
FROM tls230_appln_techn_field atf
JOIN tls901_techn_field_ipc tf ON atf.techn_field_nr = tf.techn_field_nr
JOIN tls201_appln a ON atf.appln_id = a.appln_id
WHERE a.appln_filing_year BETWEEN 2018 AND 2022
      AND atf.weight > 0.5
GROUP BY tf.techn_sector, tf.techn_field
ORDER BY application_count DESC
LIMIT 15
```

```
""")
df
```

Query took 11.00s (15 rows)

```
[28]:      techn_sector      techn_field \
0  Electrical engineering  Electrical machinery, apparatus, energy
1  Electrical engineering      Computer technology
2  Mechanical engineering      Machine tools
3      Other fields      Civil engineering
4      Instruments      Measurement
5  Mechanical engineering      Handling
6      Chemistry      Chemical engineering
7      Instruments      Medical technology
8  Mechanical engineering  Other special machines
9  Mechanical engineering      Transport
10 Mechanical engineering      Mechanical elements
11 Electrical engineering      Digital communication
12      Other fields      Furniture, games
13      Chemistry      Environmental technology
14 Electrical engineering      Audio-visual technology
```

	application_count	avg_family_size	avg_citations
0	1935413	2.087426	1.843830
1	1660916	2.514979	5.800957
2	1593869	1.332480	0.899550
3	1455691	1.480926	1.064913
4	1442095	1.755038	1.892149
5	1398741	1.551771	1.100254
6	1381834	1.448178	0.949746
7	1341127	3.110371	5.889708
8	1168549	1.693160	1.220858
9	994864	2.172360	2.196615
10	854969	1.680661	0.933609
11	800999	4.017642	7.892313
12	778959	1.830661	1.536137
13	639553	1.391839	1.016188
14	607657	3.458983	3.420591

1.4 Business Question 3

Who are the top patent applicants since 2010, and what is their patent portfolio profile? This identifies key players in the patent landscape and their innovation activity over time.

```
[29]: df = timed_query("""
SELECT
    p.doc_std_name,
    p.person_ctry_code,
```

```

COUNT(DISTINCT a.appln_id) AS total_applications,
COUNT(DISTINCT CASE WHEN a.granted = 'Y' THEN a.appln_id END) AS_
↪granted_patents,
MIN(a.appln_filing_year) AS first_filing_year,
MAX(a.appln_filing_year) AS last_filing_year,
COUNT(DISTINCT a.docdb_family_id) AS unique_patent_families
FROM tls207_pers_appln pa
JOIN tls206_person p ON pa.person_id = p.person_id
JOIN tls201_appln a ON pa.appln_id = a.appln_id
WHERE pa.applt_seq_nr > 0
      AND p.doc_std_name IS NOT NULL
      AND a.appln_filing_year >= 2010
GROUP BY p.doc_std_name, p.person_ctype_code
HAVING COUNT(DISTINCT a.appln_id) >= 50
ORDER BY total_applications DESC
LIMIT 25
""")
df

```

Query took 18.08s (25 rows)

[29]:

	doc_std_name	person_ctype_code	total_applications \
0	SAMSUNG ELECTRONICS CO LTD	KR	222459
1	STATE GRID CORP CHINA		201162
2	HUAWEI TECH CO LTD	CN	161143
3	QUALCOMM INC	US	152401
4	LG ELECTRONICS INC	KR	143595
5	BOSCH GMBH ROBERT	DE	121468
6	IBM	US	118950
7	CANON KK		108440
8	HUAWEI TECH CO LTD		97392
9	TOYOTA MOTOR CORP		91277
10	GREE ELECTRIC APPLIANCES INC ZHUHAI		90067
11	CANON KK	JP	79914
12	INTEL CORP	US	79418
13	HYUNDAI MOTOR CO LTD	KR	74280
14	IBM	KR	72643
15	APPLE INC	US	72284
16	SAMSUNG DISPLAY CO LTD	KR	72253
17	MITSUBISHI ELECTRIC CORP	JP	71651
18	TOYOTA MOTOR CO LTD	JP	68103
19	UNIV ZHEJIANG		67976
20	SIEMENS AG	DE	66372
21	HONDA MOTOR CO LTD	JP	65434
22	MICROSOFT TECHNOLOGY LICENSING LLC	US	63303
23	BOE TECHNOLOGY GROUP CO LTD		61683
24	SONY CORP	JP	60625

	granted_patents	first_filing_year	last_filing_year	\
0	105317	2010	9999	
1	129286	2010	2025	
2	54666	2010	2025	
3	61430	2010	2025	
4	80443	2010	9999	
5	31986	2010	9999	
6	89960	2010	9999	
7	54019	2010	9999	
8	54353	2010	2025	
9	46451	2010	2025	
10	53420	2010	2025	
11	53267	2010	9999	
12	42771	2010	9999	
13	41288	2010	9999	
14	50364	2010	2025	
15	48148	2010	2025	
16	44778	2010	2025	
17	28073	2010	9999	
18	37980	2010	2025	
19	42979	2010	2025	
20	24233	2010	9999	
21	41333	2010	9999	
22	33035	2010	2025	
23	38529	2010	2025	
24	32465	2010	9999	

	unique_patent_families
0	117801
1	200560
2	72418
3	47489
4	66783
5	63018
6	69091
7	93630
8	87614
9	83690
10	89625
11	49146
12	33142
13	43143
14	62032
15	29650
16	32708
17	34604

18	36885
19	67558
20	30634
21	49334
22	25842
23	60751
24	25566

1.5 Business Question 4

Which patents were most frequently cited by applications filed in 2020, and what is the typical time lag between their publication and citation? This identifies influential prior art and citation patterns.

```
[30]: df = timed_query("""
WITH citation_network AS (
    SELECT
        c.pat_publn_id,
        c.cited_pat_publn_id,
        a1.appln_id AS citing_appln_id,
        a2.appln_id AS cited_appln_id,
        a1.appln_filing_year AS citing_year,
        a2.appln_filing_year AS cited_year
    FROM tls212_citation c
    JOIN tls211_pat_publn pp1 ON c.pat_publn_id = pp1.pat_publn_id
    JOIN tls211_pat_publn pp2 ON c.cited_pat_publn_id = pp2.pat_publn_id
    JOIN tls201_appln a1 ON pp1.appln_id = a1.appln_id
    JOIN tls201_appln a2 ON pp2.appln_id = a2.appln_id
    WHERE c.cited_pat_publn_id > 0
        AND a1.appln_filing_year = 2020
)
SELECT
    cited_appln_id,
    cited_year,
    COUNT(DISTINCT citing_appln_id) AS times_cited_in_2020,
    AVG(citing_year - cited_year) AS avg_citation_lag_years
FROM citation_network
GROUP BY cited_appln_id, cited_year
HAVING COUNT(DISTINCT citing_appln_id) >= 10
ORDER BY times_cited_in_2020 DESC
LIMIT 20
""")
df
```

Query took 16.24s (20 rows)

[30]:	cited_appln_id	cited_year	times_cited_in_2020	avg_citation_lag_years
0	51239601	1983	834	37.0

1	51194878	1995	529	25.0
2	53477242	1985	523	35.0
3	472619606	2015	499	5.0
4	409574989	2012	492	8.0
5	448186713	2015	491	5.0
6	50914290	2003	464	17.0
7	53683125	1986	461	34.0
8	49656133	2006	456	14.0
9	49794663	2006	456	14.0
10	49794425	2006	452	14.0
11	49790511	2006	440	14.0
12	378600018	2011	436	9.0
13	47298897	1993	434	27.0
14	49781901	2006	433	14.0
15	50913775	2003	430	17.0
16	53424696	1991	430	29.0
17	298706670	2008	428	12.0
18	352770681	2010	425	10.0
19	469048647	2015	425	5.0

1.6 Business Question 5

How has patent activity evolved in major economies from 2015-2022, and what proportion of their patents relate to green technologies (CPC Y02 class)? This tracks innovation trends and environmental technology focus across leading countries.

```
[33]: df = timed_query("""
SELECT
    a.appln_filing_year,
    c.ctrtry_code,
    c.st3_name AS country_name,
    COUNT(DISTINCT a.appln_id) AS applications,
    COUNT(DISTINCT CASE WHEN a.granted = 'Y' THEN a.appln_id END) AS granted,
    COUNT(DISTINCT CASE WHEN cpc.cpc_class_symbol LIKE 'Y02%' THEN a.appln_id_
↪END) AS green_tech_patents,
    ROUND(COUNT(DISTINCT CASE WHEN cpc.cpc_class_symbol LIKE 'Y02%' THEN a.
↪appln_id END) * 100.0 /
        COUNT(DISTINCT a.appln_id), 2) AS green_tech_percentage
FROM tls201_appln a
JOIN tls207_pers_appln pa ON a.appln_id = pa.appln_id
JOIN tls206_person p ON pa.person_id = p.person_id
JOIN tls801_country c ON p.person_ctrtry_code = c.ctrtry_code
LEFT JOIN tls224_appln_cpc cpc ON a.appln_id = cpc.appln_id
WHERE a.appln_filing_year BETWEEN 2015 AND 2022
    AND pa.applt_seq_nr > 0
    AND c.ctrtry_code IN ('US', 'DE', 'JP', 'CN', 'KR', 'FR', 'GB')
GROUP BY a.appln_filing_year, c.ctrtry_code, c.st3_name
```

```
ORDER BY a.appln_filing_year DESC, green_tech_percentage DESC
""")
df
```

Query took 6.05s (56 rows)

```
[33]:      appln_filing_year  ctry_code      country_name  applications  \
0          2022          CN          China          169997
1          2022          KR      Republic of Korea          227764
2          2022          JP          Japan          168684
3          2022          GB      United Kingdom          32605
4          2022          FR          France          44685
5          2022          US  United States of America          390793
6          2022          DE          Germany          110946
7          2021          KR      Republic of Korea          223319
8          2021          CN          China          169150
9          2021          FR          France          44627
10         2021          GB      United Kingdom          34436
11         2021          JP          Japan          173428
12         2021          DE          Germany          117803
13         2021          US  United States of America          421462
14         2020          FR          France          46510
15         2020          KR      Republic of Korea          213784
16         2020          DE          Germany          125037
17         2020          CN          China          155938
18         2020          GB      United Kingdom          35445
19         2020          JP          Japan          179425
20         2020          US  United States of America          421681
21         2019          FR          France          49403
22         2019          DE          Germany          133154
23         2019          KR      Republic of Korea          208122
24         2019          GB      United Kingdom          35420
25         2019          JP          Japan          192285
26         2019          CN          China          132400
27         2019          US  United States of America          432975
28         2018          KR      Republic of Korea          191730
29         2018          FR          France           51132
30         2018          DE          Germany          139686
31         2018          JP          Japan          190521
32         2018          GB      United Kingdom          35238
33         2018          CN          China          116399
34         2018          US  United States of America          421596
35         2017          KR      Republic of Korea          185408
36         2017          DE          Germany          140291
37         2017          FR          France           51181
38         2017          JP          Japan          189772
39         2017          GB      United Kingdom          35620
```


40	2017	US	United States of America	431688
41	2017	CN	China	101821
42	2016	GB	United Kingdom	36014
43	2016	FR	France	53859
44	2016	KR	Republic of Korea	189183
45	2016	JP	Japan	184071
46	2016	DE	Germany	139918
47	2016	US	United States of America	430868
48	2016	CN	China	86631
49	2015	FR	France	56722
50	2015	JP	Japan	184257
51	2015	GB	United Kingdom	35460
52	2015	DE	Germany	138815
53	2015	KR	Republic of Korea	193135
54	2015	US	United States of America	432096
55	2015	CN	China	69410

	granted	green_tech_patents	green_tech_percentage
0	50420	28317	16.66
1	99014	29779	13.07
2	38700	20200	11.98
3	7848	2932	8.99
4	13200	3609	8.08
5	139027	25938	6.64
6	26975	7290	6.57
7	121454	35119	15.73
8	60628	23867	14.11
9	18267	5951	13.33
10	10990	4291	12.46
11	57154	21068	12.15
12	38253	12824	10.89
13	188627	36004	8.54
14	23099	7462	16.04
15	133323	31132	14.56
16	49325	16257	13.00
17	58260	18973	12.17
18	14576	4206	11.87
19	79183	20680	11.53
20	219796	34408	8.16
21	28379	6870	13.91
22	58172	17266	12.97
23	141745	25877	12.43
24	16853	4361	12.31
25	95474	22641	11.77
26	52615	14016	10.59
27	251741	36854	8.51
28	134935	24747	12.91

29	30620	6464	12.64
30	65630	17388	12.45
31	100436	21815	11.45
32	17874	3792	10.76
33	47654	11948	10.26
34	253387	38675	9.17
35	129590	22992	12.40
36	68479	16680	11.89
37	31011	6079	11.88
38	103406	21446	11.30
39	19036	3971	11.15
40	263301	41297	9.57
41	40423	9704	9.53
42	19066	4476	12.43
43	32742	6533	12.13
44	128844	22918	12.11
45	101650	21373	11.61
46	69648	15893	11.36
47	261333	41430	9.62
48	33178	6904	7.97
49	34150	7466	13.16
50	102299	22376	12.14
51	18788	4175	11.77
52	70698	16177	11.65
53	128003	20985	10.87
54	258979	40696	9.42
55	29121	5452	7.85

1.7 Business Question 6

Which German Federal states show the highest patent activity in Class A61B (Diagnosis/Surgery) over the last 5 years.

```
[74]: df = timed_query("""
SELECT
    n.nuts AS bundesland_code,
    n.nuts_label AS bundesland,
    COUNT(DISTINCT a.appln_id) AS total_applications,
    COUNT(DISTINCT CASE WHEN a.granted = 'Y' THEN a.appln_id END) AS
    ↪granted_patents,
    ROUND(COUNT(DISTINCT CASE WHEN a.granted = 'Y' THEN a.appln_id END) * 100.0
    ↪/
        NULLIF(COUNT(DISTINCT a.appln_id), 0), 1) AS grant_rate_pct,
    COUNT(DISTINCT p.person_id) AS unique_applicants,
    COUNT(DISTINCT a.docdb_family_id) AS unique_families

FROM tls201_appln a
```

```

JOIN tls207_pers_appln pa ON a.appln_id = pa.appln_id
JOIN tls206_person p ON pa.person_id = p.person_id
JOIN tls209_appln_ipc ipc ON a.appln_id = ipc.appln_id
JOIN tls904_nuts n ON SUBSTR(p.nuts, 1, 3) = n.nuts AND n.nuts_level = 1

WHERE a.appln_filing_year BETWEEN 2018 AND 2023
      AND pa.applt_seq_nr > 0
      AND p.person_ctry_code = 'DE'
      AND p.nuts_level >= 1
      AND ipc.ipc_class_symbol LIKE 'A61B%'

GROUP BY n.nuts, n.nuts_label
HAVING COUNT(DISTINCT a.appln_id) >= 10
ORDER BY total_applications DESC;
""")
df

```

Query took 0.52s (15 rows)

```

[74]:
bundesland_code      bundesland  total_applications  \
0                DE1      BADEN-WÜRTTEMBERG           1280
1                DE2                BAYERN             828
2                DE3                BERLIN             266
3                DE7                HESSEN             264
4                DEA      NORDRHEIN-WESTFALEN           231
5                DE6                HAMBURG            164
6                DEG                THÜRINGEN           124
7                DE9      NIEDERSACHSEN              62
8                DEF      SCHLESWIG-HOLSTEIN           62
9                DEB      RHEINLAND-PFALZ             62
10               DED                SACHSEN             38
11               DE4      BRANDENBURG                 34
12               DEE      SACHSEN-ANHALT                27
13               DE8      MECKLENBURG-VORPOMMERN         11
14               DEC                SAARLAND            11

      granted_patents  grant_rate_pct  unique_applicants  unique_families
0                  587             45.9                214             1176
1                  352             42.5                194             774
2                   75             28.2                 64             234
3                  106             40.2                 62             236
4                   76             32.9                104             208
5                   65             39.6                 29             155
6                   48             38.7                 27             118
7                   23             37.1                 35              60
8                   32             51.6                 19              60
9                   18             29.0                 21              55

```

10	11	28.9	26	36
11	17	50.0	24	33
12	3	11.1	11	24
13	4	36.4	8	9
14	4	36.4	11	11

1.8 Business Question 7 - Competitor Geographic Filing Strategy

Where are our (we are “B. Braun”) main competitors (identified by applicant name) filing their patents geographically in the medical technology field? Are they focusing on EP, US or CN filings?

STAKEHOLDER: Corporate IP Strategy / Competitive Intelligence

EXPLANATION: This query analyzes the geographic filing patterns of B. Brauns main competitors in medical technology, focusing on EP (European Patent Office), US (USPTO), and CN (CNIPA) filings. It identifies major medical device competitors and calculates their filing distribution across these key jurisdictions.

Strategic implications: - Higher filing volumes suggest strategic market prioritization - Geographic distribution reveals international expansion strategy - Comparison helps optimize own filing portfolio allocation

```
[47]: df = timed_query("""
WITH medical_tech_applications AS (
    SELECT DISTINCT
        a.appln_id,
        a.appln_auth,
        a.appln_filing_date,
        p.person_name AS applicant_name,
        p.person_id
    FROM tls201_appln a
    JOIN tls207_pers_appln pa ON a.appln_id = pa.appln_id
    JOIN tls206_person p ON pa.person_id = p.person_id
    JOIN tls230_appln_techn_field tf ON a.appln_id = tf.appln_id
    JOIN tls901_techn_field_ipc tfi ON tf.techn_field_nr = tfi.techn_field_nr
    WHERE pa.applt_seq_nr > 0
    AND tfi.techn_sector = 'Instruments'
    AND a.appln_filing_year >= 2010
    AND (
        LOWER(p.person_name) LIKE '%medtronic%' OR
        LOWER(p.person_name) LIKE '%johnson%johnson%' OR
        LOWER(p.person_name) LIKE '%abbott%' OR
        LOWER(p.person_name) LIKE '%boston%scientific%' OR
        LOWER(p.person_name) LIKE '%stryker%' OR
        LOWER(p.person_name) LIKE '%zimmer%' OR
        LOWER(p.person_name) LIKE '%smith%nephew%' OR
        LOWER(p.person_name) LIKE '%edwards%' OR
        LOWER(p.person_name) LIKE '%baxter%' OR
        LOWER(p.person_name) LIKE '%fresenius%' OR
```

```

        LOWER(p.person_name) LIKE '%braun%'
    )
)

SELECT
    mta.applicant_name,
    mta.appln_auth AS filing_authority,
    COUNT(*) AS patent_count,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (PARTITION BY mta.
↪applicant_name), 2) AS percentage_by_applicant,
    CASE
        WHEN mta.appln_auth = 'EP' THEN 'European Patent Office'
        WHEN mta.appln_auth = 'US' THEN 'United States Patent Office'
        WHEN mta.appln_auth = 'CN' THEN 'China Patent Office'
        ELSE 'Other Authority'
    END AS authority_description
FROM medical_tech_applications mta
WHERE mta.appln_auth IN ('EP', 'US', 'CN')
GROUP BY mta.applicant_name, mta.appln_auth
HAVING COUNT(*) >= 10
ORDER BY mta.applicant_name, patent_count DESC;
""")
df

```

Query took 4.90s (354 rows)

```

[47]:
      applicant_name  filing_authority  patent_count  \
0      ABBOTT CARDIOVASCULAR SYSTEMS      CN          15
1      ABBOTT CARDIOVASCULAR SYSTEMS INC.      CN          20
2      ABBOTT CARDIOVASCULAR SYSTEMS INC.      US          12
3      ABBOTT DIABETES CARE INC.      CN          59
4      ABBOTT LAB      CN          18
..      ...      ...      ...
349      Zimmer US, Inc.      US          10
350      Zimmer, Inc.      US          728
351      Zimmer, Inc.      EP          267
352      Zimmer, Juergen      US          11
353      Zimmerling, Martin      US          16

      percentage_by_applicant  authority_description
0          100.00      China Patent Office
1           62.50      China Patent Office
2           37.50  United States Patent Office
3          100.00      China Patent Office
4          100.00      China Patent Office
..      ...      ...
349          100.00  United States Patent Office

```

350	73.17	United States Patent Office
351	26.83	European Patent Office
352	100.00	United States Patent Office
353	100.00	United States Patent Office

[354 rows x 5 columns]

1.9 Business Question 8 - Diagnostic Imaging Grant Rates by Patent Office

What is the grant rate for diagnostic imaging patents (A61B 6/) at EPO vs. USPTO vs. CNIPA, and how does this affect our international filing strategy?

STAKEHOLDER: Patent Prosecution / IP Portfolio Management

EXPLANATION: This query analyzes grant rates for diagnostic imaging patents (IPC subclass A61B 6/) across three major patent offices. A61B 6/ covers diagnostic imaging technologies including X-ray, ultrasound, MRI, and other medical imaging devices.

IPC NOTE: FIXED - Uses combined LIKE patterns to handle variable whitespace Pattern: LIKE 'A61B%' AND LIKE '%6/%' with exclusions for 16/, 26/, 36/, 46/, 56/

Strategic implications: - Higher grant rates suggest more favorable examination practices - Lower grant rates may indicate stricter standards but stronger patents - Application volumes show market activity and competitive landscape - Data helps inform first-filing decisions and budget planning

```
[79]: df = timed_query("""
WITH diagnostic_imaging_patents AS (
    SELECT DISTINCT
        a.appln_id,
        a.appln_auth,
        a.granted
    FROM tls201_appln a
    JOIN tls209_appln_ipc ipc ON a.appln_id = ipc.appln_id
    WHERE ipc.ipc_class_symbol LIKE 'A61B%'
        AND ipc.ipc_class_symbol LIKE '%6/%'
        AND ipc.ipc_class_symbol NOT LIKE '%16/%'
        AND ipc.ipc_class_symbol NOT LIKE '%46/%'
        AND a.appln_auth IN ('EP', 'US', 'CN')
        AND a.appln_filing_year BETWEEN 2010 AND 2023
)

SELECT
    CASE
        WHEN appln_auth = 'EP' THEN 'European Patent Office (EPO)'
        WHEN appln_auth = 'US' THEN 'USPTO'
        WHEN appln_auth = 'CN' THEN 'CNIPA'
    END AS office_name,
    appln_auth AS office_code,
    COUNT(*) AS total_applications,
```

```

COUNT(CASE WHEN granted = 'Y' THEN 1 END) AS granted_patents,
ROUND(COUNT(CASE WHEN granted = 'Y' THEN 1 END) * 100.0 / COUNT(*), 1) AS_
↪grant_rate_pct
FROM diagnostic_imaging_patents
GROUP BY appln_auth
ORDER BY grant_rate_pct DESC;
""")
df

```

Query took 0.62s (3 rows)

```

[79]:
      office_name office_code total_applications \
0             USPTO         US             21446
1             CNIPA         CN             32674
2  European Patent Office (EPO)         EP             7144

      granted_patents  grant_rate_pct
0             16957             79.1
1             23873             73.1
2             3837             53.7

```

1.10 Business Question 9 - AI-based Enterprise Resource Planning Patent Landscape

How has the patent landscape for AI-based enterprise resource planning (G06Q10 + G06N) evolved since 2018, and who are the top 10 applicants we should monitor?

STAKEHOLDER: Technology Scouting / Strategic Planning

EXPLANATION: This query analyzes the patent landscape for AI-based ERP by identifying applications with both G06Q10 (ERP/business methods) and G06N (AI/machine learning) CPC classifications since 2018.

CPC NOTE: FIXED - CPC codes also have variable whitespace Using pattern 'G06Q%' AND '%10/%' for G06Q 10/ Using pattern 'G06N%' for main class (all G06N subclasses)

The query provides: - Yearly filing trends with year-over-year growth - Top 10 most active applicants with their portfolio characteristics - Geographic distribution and sector classification of key players

```

[80]: df = timed_query("""
WITH ai_erp_patents AS (
  SELECT DISTINCT
    a.appln_id,
    a.appln_filing_date,
    a.appln_filing_year
  FROM tls201_appln a
  WHERE a.appln_filing_year >= 2018
  AND EXISTS (
    SELECT 1 FROM tls224_appln_cpc cpc1
    WHERE cpc1.appln_id = a.appln_id

```

```

        AND cpc1.cpc_class_symbol LIKE 'G06Q%'
    )
    AND EXISTS (
        SELECT 1 FROM tls224_appln_cpc cpc2
        WHERE cpc2.appln_id = a.appln_id
        AND cpc2.cpc_class_symbol LIKE 'G06N%'
    )
),

yearly_trends AS (
    SELECT
        appln_filing_year,
        COUNT(*) AS patent_count,
        COUNT(*) - LAG(COUNT(*)) OVER (ORDER BY appln_filing_year) AS_
↪year_over_year_change
    FROM ai_erp_patents
    GROUP BY appln_filing_year
),

top_applicants AS (
    SELECT
        p.person_id,
        p.person_name,
        p.person_ctype_code,
        p.psn_sector,
        COUNT(DISTINCT aep.appln_id) AS patent_count,
        MIN(aep.appln_filing_date) AS first_filing_date,
        MAX(aep.appln_filing_date) AS latest_filing_date,
        COUNT(DISTINCT aep.appln_filing_year) AS active_years
    FROM ai_erp_patents aep
    JOIN tls207_pers_appln pa ON aep.appln_id = pa.appln_id
    JOIN tls206_person p ON pa.person_id = p.person_id
    WHERE pa.applt_seq_nr > 0
    GROUP BY p.person_id, p.person_name, p.person_ctype_code, p.psn_sector
    ORDER BY patent_count DESC
    LIMIT 10
)

-- Yearly Trends
SELECT
    'YEARLY_TRENDS' AS result_type,
    CAST(appln_filing_year AS STRING) AS identifier,
    NULL AS person_name,
    NULL AS country,
    NULL AS sector,
    patent_count,
    year_over_year_change AS change_or_years,

```



```

        NULL AS first_filing,
        NULL AS latest_filing
FROM yearly_trends

UNION ALL

-- Top Applicants
SELECT
    'TOP_APPLICANTS' AS result_type,
    CAST(ROW_NUMBER() OVER (ORDER BY patent_count DESC) AS STRING) AS_
⇨identifier,
    person_name,
    person_ctry_code AS country,
    psn_sector AS sector,
    patent_count,
    active_years AS change_or_years,
    CAST(first_filing_date AS STRING) AS first_filing,
    CAST(latest_filing_date AS STRING) AS latest_filing
FROM top_applicants

ORDER BY result_type, identifier;
""")
df

```

Query took 0.55s (18 rows)

```

[80]:      result_type identifier \
0    TOP_APPLICANTS          1
1    TOP_APPLICANTS         10
2    TOP_APPLICANTS          2
3    TOP_APPLICANTS          3
4    TOP_APPLICANTS          4
5    TOP_APPLICANTS          5
6    TOP_APPLICANTS          6
7    TOP_APPLICANTS          7
8    TOP_APPLICANTS          8
9    TOP_APPLICANTS          9
10   YEARLY_TRENDS        2018
11   YEARLY_TRENDS        2019
12   YEARLY_TRENDS        2020
13   YEARLY_TRENDS        2021
14   YEARLY_TRENDS        2022
15   YEARLY_TRENDS        2023
16   YEARLY_TRENDS        2024
17   YEARLY_TRENDS        2025

```

```

person_name country      sector \

```

0	STATE GRID CORPORATION OF CHINA	COMPANY
1	NORTH CHINA ELECTRIC POWER UNIVERSITY	UNIVERSITY
2	INDUSTRIAL AND COMMERCIAL BANK OF CHINA CO., LTD.	COMPANY
3	Capital One Services, LLC	US COMPANY
4	International Business Machines Corporation	US COMPANY
5	GUANGDONG POWER GRID COMPANY	COMPANY
6	Bank of China Co., Ltd.	COMPANY
7	CHINA ELECTRIC POWER RESEARCH INSTITUTE CO., LTD.	COMPANY
8	ZHEJIANG UNIVERSITY	UNIVERSITY
9	SOUTHEAST UNIVERSITY	UNIVERSITY
10	None	None
11	None	None
12	None	None
13	None	None
14	None	None
15	None	None
16	None	None
17	None	None

	patent_count	change_or_years	first_filing	latest_filing
0	2135	8.0	2018-04-09	2025-05-20
1	502	8.0	2018-01-04	2025-05-14
2	982	7.0	2019-06-19	2025-03-11
3	976	8.0	2018-01-04	2025-04-22
4	933	7.0	2018-01-02	2024-02-15
5	648	8.0	2018-04-08	2025-05-06
6	600	7.0	2018-03-06	2024-11-12
7	577	8.0	2018-01-18	2025-04-30
8	573	8.0	2018-03-12	2025-05-26
9	513	8.0	2018-01-19	2025-05-09
10	8271	NaN	None	None
11	12720	4449.0	None	None
12	18478	5758.0	None	None
13	20539	2061.0	None	None
14	19182	-1357.0	None	None
15	22570	3388.0	None	None
16	28748	6178.0	None	None
17	5636	-23112.0	None	None

1.11 Business Question 10 - German Federal States - Medical Diagnosis/Surgery Patent Activity

Which German Federal states show the highest patent activity in Class A61B (Diagnosis/Surgery) over the last 5 years, and how does our region compare to the national average of patents per mio inhabitants.

STAKEHOLDER: Regional PATLIB Center / Economic Development Agency

EXPLANATION: This query analyzes patent activity in IPC class A61B (medical diagno-

sis/surgery) over the last 5 years, focusing on German federal states with comparison to European countries.

IPC NOTE: Uses main class A61B% only - simple pattern works fine

Key outputs: - German federal states ranked by patent count - Deviation from national average (%) - European country comparison for context - Both patent and applicant counts per region

```
[81]: df = timed_query("""
WITH population_2023 AS (
    -- Quelle: Statistisches Bundesamt (destatis.de), Stand 31.12.2023
    SELECT * FROM UNNEST([
        STRUCT('DE1' AS nuts_code, 'BADEN-WÜRTTEMBERG' AS bundesland, 11280000_
        ↪AS population),
        STRUCT('DE2', 'BAYERN', 13370000),
        STRUCT('DE3', 'BERLIN', 3850000),
        STRUCT('DE4', 'BRANDENBURG', 2570000),
        STRUCT('DE5', 'BREMEN', 680000),
        STRUCT('DE6', 'HAMBURG', 1950000),
        STRUCT('DE7', 'HESSEN', 6390000),
        STRUCT('DE8', 'MECKLENBURG-VORPOMMERN', 1610000),
        STRUCT('DE9', 'NIEDERSACHSEN', 8140000),
        STRUCT('DEA', 'NORDRHEIN-WESTFALEN', 18140000),
        STRUCT('DEB', 'RHEINLAND-PFALZ', 4160000),
        STRUCT('DEC', 'SAARLAND', 990000),
        STRUCT('DED', 'SACHSEN', 4090000),
        STRUCT('DEE', 'SACHSEN-ANHALT', 2170000),
        STRUCT('DEF', 'SCHLESWIG-HOLSTEIN', 2950000),
        STRUCT('DEG', 'THÜRINGEN', 2110000)
    ])
),

recent_patents AS (
    SELECT
        a.appln_id,
        p.person_id,
        p.nuts
    FROM tls201_appln a
    JOIN tls209_appln_ipc ipc ON a.appln_id = ipc.appln_id
    JOIN tls207_pers_appln pa ON a.appln_id = pa.appln_id
    JOIN tls206_person p ON pa.person_id = p.person_id
    WHERE ipc.ipc_class_symbol LIKE 'A61B%'
        AND a.appln_filing_year >= 2019
        AND pa.applt_seq_nr > 0
        AND p.person_ctype_code = 'DE'
        AND p.nuts_level >= 1
),
```

```

german_state_activity AS (
    SELECT
        SUBSTR(rp.nuts, 1, 3) AS nuts_code,
        COUNT(DISTINCT rp.appln_id) AS patent_count,
        COUNT(DISTINCT rp.person_id) AS applicant_count
    FROM recent_patents rp
    GROUP BY SUBSTR(rp.nuts, 1, 3)
)

SELECT
    RANK() OVER (ORDER BY gsa.patent_count DESC) AS rank_total,
    pop.nuts_code,
    pop.bundesland,
    gsa.patent_count,
    gsa.applicant_count,
    ROUND(pop.population / 1000000.0, 2) AS population_mio,
    ROUND(gsa.patent_count * 1000000.0 / pop.population, 1) AS
    ↪patents_per_mio_inhabitants,
    RANK() OVER (ORDER BY gsa.patent_count * 1.0 / pop.population DESC) AS
    ↪rank_per_capita,
    ROUND(gsa.patent_count * 100.0 / SUM(gsa.patent_count) OVER (), 1) AS
    ↪pct_of_total
FROM german_state_activity gsa
JOIN population_2023 pop ON gsa.nuts_code = pop.nuts_code
ORDER BY patents_per_mio_inhabitants DESC;
""")
df

```

Query took 0.55s (16 rows)

```

[81]:
   rank_total  nuts_code  bundesland  patent_count  \
0           1      DE1    BADEN-WÜRTTEMBERG      1181
1           6      DE6           HAMBURG        147
2           4      DE3           BERLIN        238
3           2      DE2           BAYERN        682
4           7      DEG           THÜRINGEN       104
5           3      DE7           HESSEN        239
6           9      DEF    SCHLESWIG-HOLSTEIN        57
7           8      DEB    RHEINLAND-PFALZ         60
8          12      DE4           BRANDENBURG        30
9          13      DEE    SACHSEN-ANHALT         25
10          5      DEA    NORDRHEIN-WESTFALEN       187
11          15      DEC           SAARLAND         8
12          11      DED           SACHSEN         33
13          14      DE8    MECKLENBURG-VORPOMMERN       11
14          10      DE9           NIEDERSACHSEN       52
15          16      DE5           BREMEN         3

```

	applicant_count	population_mio	patents_per_mio_inhabitants	\
0	196	11.28	104.7	
1	24	1.95	75.4	
2	57	3.85	61.8	
3	164	13.37	51.0	
4	25	2.11	49.3	
5	57	6.39	37.4	
6	19	2.95	19.3	
7	21	4.16	14.4	
8	22	2.57	11.7	
9	11	2.17	11.5	
10	91	18.14	10.3	
11	8	0.99	8.1	
12	26	4.09	8.1	
13	8	1.61	6.8	
14	30	8.14	6.4	
15	3	0.68	4.4	

	rank_per_capita	pct_of_total
0	1	38.6
1	2	4.8
2	3	7.8
3	4	22.3
4	5	3.4
5	6	7.8
6	7	1.9
7	8	2.0
8	9	1.0
9	10	0.8
10	11	6.1
11	12	0.3
12	13	1.1
13	14	0.4
14	15	1.7
15	16	0.1

1.12 Business Question 11 - Fast growing IT methods for management patents

What are the fastest-growing sub-classes within G06Q (IT methods for management) in the last 3 years, and which applicants are driving this growth?

STAKEHOLDER: Technology Transfer Office / Innovation Scouting

EXPLANATION: This query identifies the fastest-growing G06Q sub-classes (IT methods for business/management) by comparing filing activity between the base year (3 years ago) and the most recent year.

Key outputs: - Top 10 fastest-growing subclasses by growth rate - Top 3 applicants driving growth

in each subclass - Filing counts and growth percentages */

```
[75]: df = timed_query("""

WITH g06q_subclasses AS (
    SELECT
        SUBSTR(cpc.cpc_class_symbol, 1, 8) AS subclass,
        a.appln_id,
        a.appln_filing_date,
        a.appln_filing_year
    FROM tls224_appln_cpc cpc
    JOIN tls201_appln a ON cpc.appln_id = a.appln_id
    WHERE cpc.cpc_class_symbol LIKE 'G06Q%'
        AND a.appln_filing_year >= 2021 -- Last 3 years
),

subclass_growth AS (
    SELECT
        subclass,
        COUNT(*) AS total_applications,
        COUNT(CASE WHEN appln_filing_year = 2023 THEN 1 END) AS
↪recent_year_apps,
        COUNT(CASE WHEN appln_filing_year = 2021 THEN 1 END) AS base_year_apps,
        CASE
            WHEN COUNT(CASE WHEN appln_filing_year = 2021 THEN 1 END) > 0
            THEN (CAST(COUNT(CASE WHEN appln_filing_year = 2023 THEN 1 END) AS
↪FLOAT64) /
            COUNT(CASE WHEN appln_filing_year = 2021 THEN 1 END) - 1) *
↪100
            ELSE NULL
        END AS growth_rate_pct
    FROM g06q_subclasses
    GROUP BY subclass
    HAVING COUNT(*) >= 10
),

top_growing_subclasses AS (
    SELECT *
    FROM subclass_growth
    WHERE growth_rate_pct IS NOT NULL
    ORDER BY growth_rate_pct DESC
    LIMIT 10
),

driving_applicants AS (
    SELECT
        tgs.subclass,
```

```

        tgs.growth_rate_pct,
        p.person_name,
        p.person_ctype_code,
        COUNT(*) AS applications_count,
        ROW_NUMBER() OVER (PARTITION BY tgs.subclass ORDER BY COUNT(*) DESC) AS_
↪applicant_rank
    FROM top_growing_subclasses tgs
    JOIN g06q_subclasses gqs ON tgs.subclass = gqs.subclass
    JOIN t1s207_pers_appln pa ON gqs.appln_id = pa.appln_id
    JOIN t1s206_person p ON pa.person_id = p.person_id
    WHERE pa.applt_seq_nr > 0
    GROUP BY tgs.subclass, tgs.growth_rate_pct, p.person_name, p.
↪person_ctype_code
)

SELECT
    da.subclass,
    ROUND(da.growth_rate_pct, 2) AS growth_rate_percent,
    da.person_name AS top_applicant,
    da.person_ctype_code AS applicant_country,
    da.applications_count AS applicant_filings,
    tgs.total_applications AS subclass_total_apps,
    tgs.recent_year_apps,
    tgs.base_year_apps
FROM driving_applicants da
JOIN top_growing_subclasses tgs ON da.subclass = tgs.subclass
WHERE da.applicant_rank <= 3
ORDER BY da.growth_rate_pct DESC, da.subclass, da.applicant_rank;
""")
df

```

Query took 5.88s (30 rows)

```

[75]:      subclass  growth_rate_percent  \
0    G06Q  50          10.78
1    G06Q  50          10.78
2    G06Q  50          10.78
3    G06Q  10           0.36
4    G06Q  10           0.36
5    G06Q  10           0.36
6    G06Q  40          -2.85
7    G06Q  40          -2.85
8    G06Q  40          -2.85
9    G06Q2220         -3.70
10   G06Q2220         -3.70
11   G06Q2220         -3.70
12   G06Q  30         -16.64

```

13	G06Q	30	-16.64
14	G06Q	30	-16.64
15	G06Q2230		-19.32
16	G06Q2230		-19.32
17	G06Q2230		-19.32
18	G06Q	20	-27.16
19	G06Q	20	-27.16
20	G06Q	20	-27.16
21	G06Q	99	-27.73
22	G06Q	99	-27.73
23	G06Q	99	-27.73
24	G06Q	90	-27.97
25	G06Q	90	-27.97
26	G06Q	90	-27.97
27	G06Q2240		-43.14
28	G06Q2240		-43.14
29	G06Q2240		-43.14

	top_applicant	applicant_country	\
0	STATE GRID CORPORATION OF CHINA		
1	GUANGDONG POWER GRID COMPANY		
2	CHINA ELECTRIC POWER RESEARCH INSTITUTE CO., LTD.		
3	COUPANG CORP.	KR	
4	STATE GRID CORPORATION OF CHINA		
5	GUANGDONG POWER GRID COMPANY		
6	INDUSTRIAL AND COMMERCIAL BANK OF CHINA CO., LTD.		
7	Bank of China Co., Ltd.		
8	CHINA CONSTRUCTION BANK CORPORATION		
9	Capital One Services, LLC	US	
10	BASF SE	DE	
11	Visa International Service Association	US	
12	COUPANG CORP.	KR	
13	Capital One Services, LLC	US	
14	GOOGLE LLC	US	
15	Efficient Auctions LLC	US	
16	FANMOUNTAIN LLC	US	
17	iCashe, Inc.	US	
18	Capital One Services, LLC	US	
19	Visa International Service Association	US	
20	CAPITAL ONE SERVICES, LLC	US	
21	Strong Force EE Portfolio 2022, LLC	US	
22	Middle Chart, LLC	US	
23	EMOTIONAL PERCEPTION AI LIMITED	GB	
24	TRAX TECHNOLOGY SOLUTIONS PTE LTD.	SG	
25	LG Electronics Inc.	KR	
26	LGHorizon, LLC	US	
27	Toyota Jidosha Kabushiki Kaisha	JP	

28	KIA CORPORATION	KR
29	Hyundai Motor Company	KR

	applicant_filings	subclass_total_apps	recent_year_apps	base_year_apps
0	6259	298965	76232	68816
1	2621	298965	76232	68816
2	1696	298965	76232	68816
3	8571	491475	120495	120060
4	6831	491475	120495	120060
5	2901	491475	120495	120060
6	3425	71556	18827	19380
7	3256	71556	18827	19380
8	2106	71556	18827	19380
9	153	9382	2394	2486
10	116	9382	2394	2486
11	115	9382	2394	2486
12	8125	248600	61151	73361
13	1580	248600	61151	73361
14	1489	248600	61151	73361
15	6	277	71	88
16	4	277	71	88
17	4	277	71	88
18	6591	191391	45606	62613
19	4017	191391	45606	62613
20	3562	191391	45606	62613
21	32	317	86	119
22	32	317	86	119
23	10	317	86	119
24	15	774	170	236
25	13	774	170	236
26	12	774	170	236
27	39	844	170	299
28	35	844	170	299
29	19	844	170	299

1.13 Business Question 12 - AI assisted diagnostics companies

Which companies are building patent portfolios in AI-assisted diagnostics (intersection of A61B + G06N), and what is the average time-to-grant?

STAKEHOLDER: Competitive Intelligence / R&D Strategy

EXPLANATION: This query identifies companies active in AI-assisted diagnostics by finding patents at the intersection of medical diagnosis (A61B) and artificial intelligence (G06N) classifications.

IPC NOTE: Uses main classes A61B% and G06N% - simple patterns work fine

```

[76]: df = timed_query("""
WITH ai_diagnostics_patents AS (
    SELECT DISTINCT
        a61b.appln_id,
        app.appln_filing_date,
        app.granted,
        app.earliest_filing_date
    FROM tls209_appln_ipc a61b
    JOIN tls209_appln_ipc g06n ON a61b.appln_id = g06n.appln_id
    JOIN tls201_appln app ON a61b.appln_id = app.appln_id
    WHERE a61b.ipc_class_symbol LIKE 'A61B%'
        AND g06n.ipc_class_symbol LIKE 'G06N%'
        AND app.granted = 'Y'
),

granted_patents_with_events AS (
    SELECT
        adp.appln_id,
        adp.appln_filing_date,
        adp.earliest_filing_date,
        le.event_effective_date AS grant_date
    FROM ai_diagnostics_patents adp
    JOIN tls231_inpadoc_legal_event le ON adp.appln_id = le.appln_id
    WHERE le.event_type = 'G'
        AND le.event_effective_date IS NOT NULL
),

company_patents AS (
    SELECT
        gpe.appln_id,
        gpe.appln_filing_date,
        gpe.earliest_filing_date,
        gpe.grant_date,
        p.person_id,
        p.person_name,
        p.psn_sector
    FROM granted_patents_with_events gpe
    JOIN tls207_pers_appln pa ON gpe.appln_id = pa.appln_id
    JOIN tls206_person p ON pa.person_id = p.person_id
    WHERE pa.applt_seq_nr > 0
        AND p.psn_sector = 'COMPANY'
),

time_to_grant_calc AS (
    SELECT
        person_id,
        person_name,

```

```

        appln_id,
        appln_filing_date,
        grant_date,
        DATE_DIFF(grant_date, appln_filing_date, DAY) AS days_to_grant
    FROM company_patents
    WHERE grant_date IS NOT NULL
        AND appln_filing_date IS NOT NULL
        AND grant_date >= appln_filing_date
)

SELECT
    person_name AS company_name,
    COUNT(DISTINCT appln_id) AS patent_count,
    ROUND(AVG(days_to_grant), 0) AS avg_days_to_grant,
    ROUND(AVG(days_to_grant) / 365.25, 1) AS avg_years_to_grant
FROM time_to_grant_calc
GROUP BY person_id, person_name
HAVING COUNT(DISTINCT appln_id) >= 2
ORDER BY patent_count DESC, avg_days_to_grant ASC;
""")
df

```

Query took 4.00s (0 rows)

```

[76]: Empty DataFrame
      Columns: []
      Index: []

```

1.14 Business Question 13 - Regional Patent Comparison by Technology Sector

How does patent activity in our federal state (e.g., Sachsen) compare to peer regions like Bavaria or Baden-Württemberg - broken down by technology sectors?

STAKEHOLDER: Regional Economic Development / PATLIB Center

EXPLANATION: This query compares patent activity across German federal states (Sachsen, Bayern, Baden-Württemberg) broken down by WIPO technology sectors.

Key outputs: - Patent counts by region and technology sector - Recent patents (2018+) to show current activity - Average technology relevance weight - Time span of patent activity per region-sector combination

Note: Uses NUTS level 1 for federal state identification and joins with technology field classification tables.

```

[77]: df = timed_query("""
      WITH regional_patents AS (
        SELECT
          a.appln_id,
          a.appln_filing_year,

```

```

        p.nuts,
        n.nuts_label,
        CASE
            WHEN LOWER(n.nuts_label) LIKE '%sachsen%' OR LOWER(n.nuts_label) LIKE
↪LIKE '%saxony%' THEN 'Sachsen'
            WHEN LOWER(n.nuts_label) LIKE '%bayern%' OR LOWER(n.nuts_label) LIKE
↪LIKE '%bavaria%' THEN 'Bayern'
            WHEN LOWER(n.nuts_label) LIKE '%baden%württemberg%' OR LOWER(n.
↪nuts_label) LIKE '%baden-württemberg%' THEN 'Baden-Württemberg'
            ELSE 'Other'
        END AS region_group
FROM tls201_appln a
JOIN tls207_pers_appln pa ON a.appln_id = pa.appln_id
JOIN tls206_person p ON pa.person_id = p.person_id
JOIN tls904_nuts n ON p.nuts = n.nuts
WHERE pa.applt_seq_nr > 0
      AND a.appln_filing_year >= 2010
      AND n.nuts_level = 1
      AND (LOWER(n.nuts_label) LIKE '%sachsen%' OR LOWER(n.nuts_label) LIKE
↪'%saxony%'
           OR LOWER(n.nuts_label) LIKE '%bayern%' OR LOWER(n.nuts_label) LIKE
↪'%bavaria%'
           OR LOWER(n.nuts_label) LIKE '%baden%württemberg%' OR LOWER(n.
↪nuts_label) LIKE '%baden-württemberg%')
),

technology_sectors AS (
    SELECT
        rp.appln_id,
        rp.region_group,
        rp.appln_filing_year,
        tf.techn_sector,
        tf.techn_field,
        atf.weight
    FROM regional_patents rp
    JOIN tls230_appln_techn_field atf ON rp.appln_id = atf.appln_id
    JOIN tls901_techn_field_ipc tf ON atf.techn_field_nr = tf.techn_field_nr
    WHERE rp.region_group != 'Other'
)

SELECT
    region_group,
    techn_sector,
    COUNT(DISTINCT appln_id) AS patent_count,
    COUNT(DISTINCT CASE WHEN appln_filing_year >= 2018 THEN appln_id END) AS
↪recent_patents_2018_plus,

```

```
        ROUND(AVG(weight), 3) AS avg_tech_relevance_weight,  
        MIN(appln_filing_year) AS earliest_year,  
        MAX(appln_filing_year) AS latest_year  
FROM technology_sectors  
GROUP BY region_group, techn_sector  
ORDER BY region_group, patent_count DESC;  
""")  
df
```

Query took 3.59s (0 rows)

```
[77]: Empty DataFrame  
      Columns: []  
      Index: []
```